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Studierichting: Informatica
Oefeningenreeks 4A nr. 51.11

$$\text{Opgave: } I = \int (\sin^2 x \cos^3 x) dx$$

$$= \int (\sin^2 x \cos^2 x \cos x) dx$$

$$= \int (\sin^2 x \cos^2 x) d(\sin x)$$

$$= \int (\sin^2 x (1 - \sin^2 x)) d(\sin x)$$

$$\Rightarrow t = \sin x$$

$$I = \int t^2 (1 - t^2) dt = \int (t^2 - t^4) dt = \int t^2 dt - \int t^4 dt$$

$$= \frac{t^3}{3} - \frac{t^5}{5} + c = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + c$$

References